

# SOFTWARE ARCHITECTURE AND DESIGN

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(CMP-N207-0)

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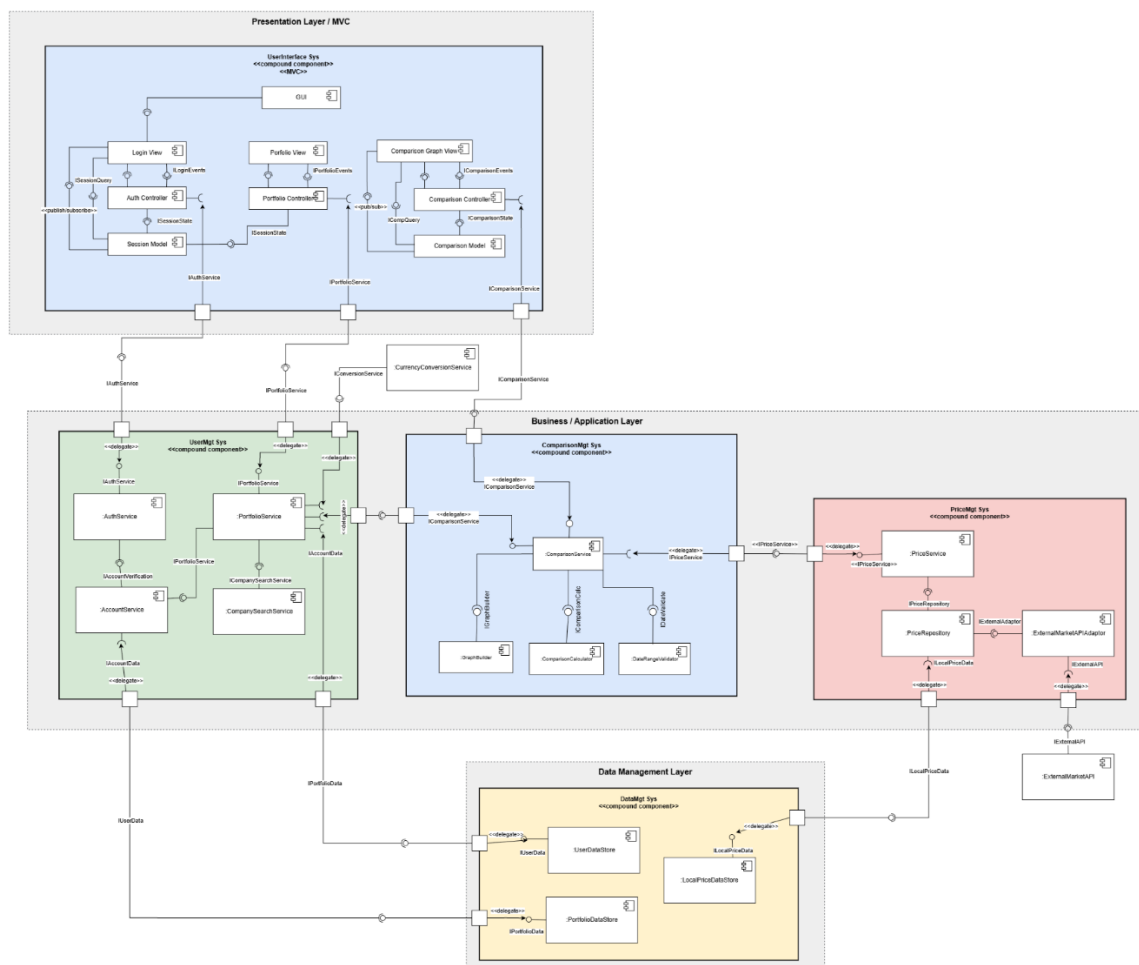
# Introduction:

This report presents the final implementation of the share price comparison web application we have been working on this semester. Sprint 3 focuses on refining the system architecture by introducing compound components, applying architectural styles and implementing service-oriented architecture followed by system testing.

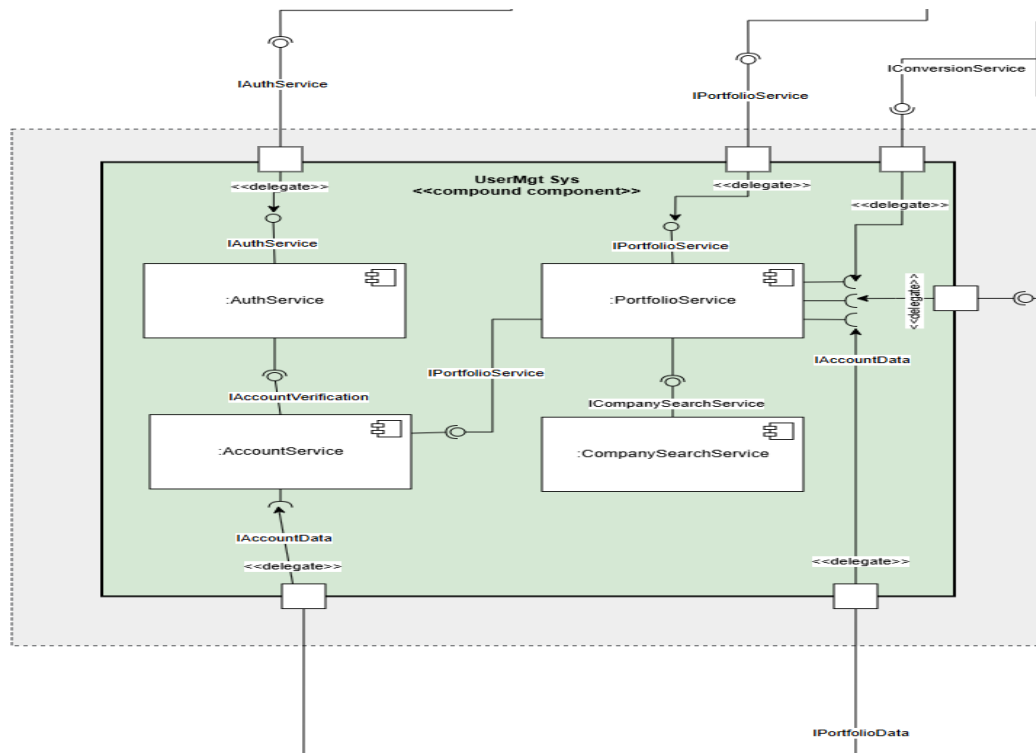
## Objectives:

- Implement compound components
- Apply architectural styles
- Introduce SOA principles
- Complete system implementation
- Test the system

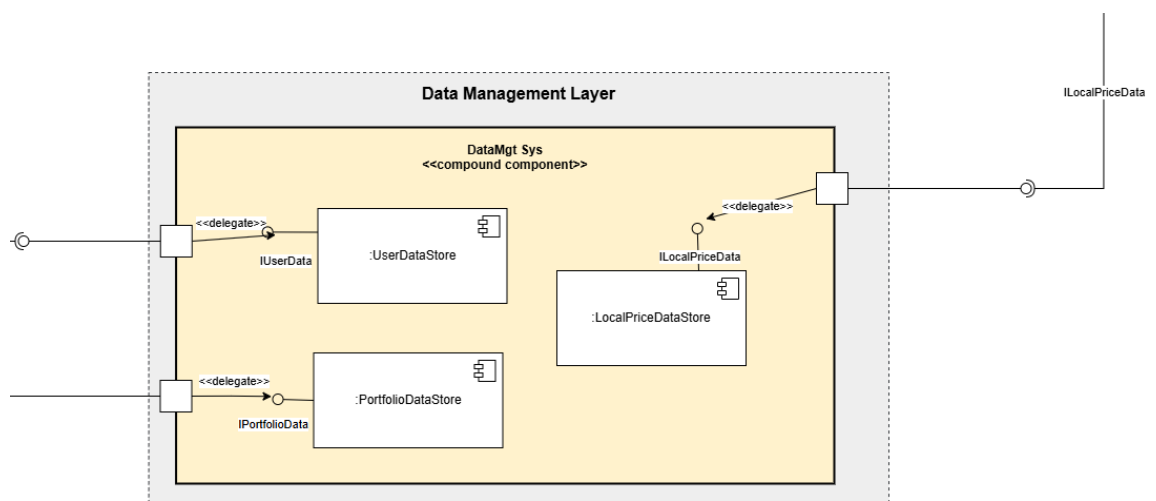
## Final System Architecture:



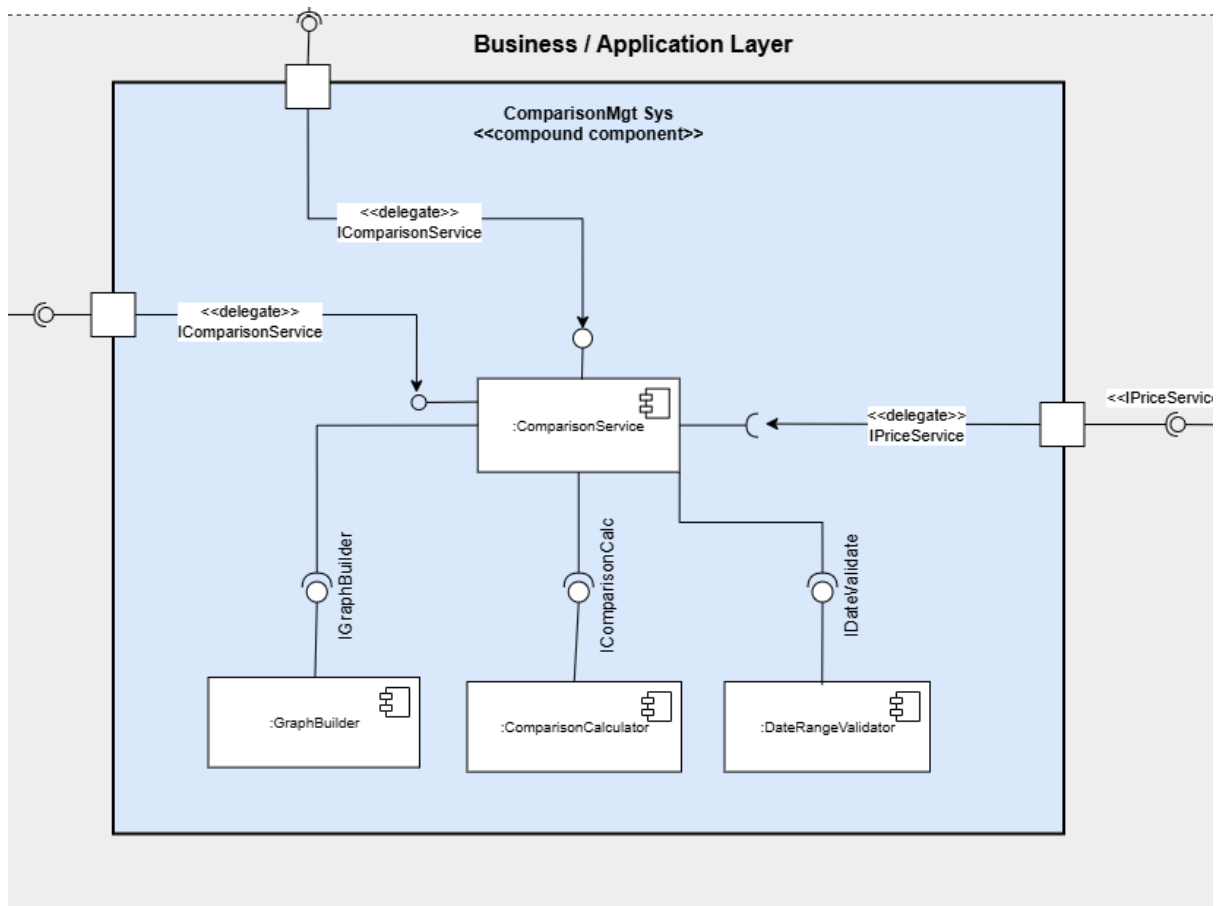
The system is divided into 4 main parts, the UserMgt Sys, DataMgt Sys, PriceMgt Sys, and ComparisonMGT Sys. Each part has and handles a specific role in our web comparison app which follow:



- User-related functions are managed by the UserMgt Sys. This covers currency conversion, portfolio management, company research, account administration, and authentication. User operations are divided into smaller, reusable services by services like AuthService, AccountService, PortfolioService, and CompanySearchService.







- The application's comparison functionalities are managed by the ComparisonMgt Sys. It consists of the DateRangeValidator, ComparisonCalculator, GraphBuilder, and ComparisonService. This system compiles the data required to produce graphs, verifies the chosen date range, compares share values between firms, and computes comparison results.

The system follows a service oriented and layered architecture where each part works independently but communicates through clear interfaces which improves flexibility, scalability and maintainability.

## Architectural style:

### 3 Layer Architecture:

The system follows a layered architecture where components are organised into separate layers, each with a specific responsibility. According to the lecture material, layered architecture structures systems hierarchically, where each layer communicates only with adjacent layers through defined interfaces

The system is divided into the presentation Layer, Business Layer and Data Layer:

**Presentation Layer (UserMgt Sys)**

Handles user interaction, including authentication, account management, and portfolio operations.

**Business Logic Layer (ComparisonMgt Sys & PriceMgt Sys)**

Processes data and performs core application logic such as comparing share prices, validating input, and retrieving market data.

**Data Layer (DataMgt Sys)**

Responsible for storing and managing data, including user data, portfolio data, and locally stored share prices.

**Service-Oriented Architecture (SOA)**

The system also follows a Service-Oriented Architecture (SOA), where functionality is divided into independent services that communicate with each other.

In this system Services such as:

- authService
- AccountService
- PortfolioService
- ComparisonService
- MarketDataProvider

Are designed as independent units each responsible for a specific function.

Furthermore the services communicate through interfaces, reducing dependency between components which allows the system to be more efficient and flexible and part can be updated or replace without affecting the whole system.

## Domain Independent Style

**Adapter Pattern**

The Adapter pattern is used within the PriceMgt Sys, specifically through the MarketDataProvider.

As described in the lecture slides in week 9 , the Adapter pattern is used when a system needs to interact with an external component that provides a different interface .

In this system, the MarketDataProvider acts as an adapter between the application and the external market API. This ensures that changes to the external API do not affect the rest of the system.

## Design rationale

To enhance organization, scalability, and maintainability, the system was created with compound components, layered architecture, and service-oriented principles. Compound components make the system easier to administer by grouping related functions into subsystems. The presentation, business logic, and data layers are kept apart by the three-layer design, which guarantees a distinct division of responsibilities. Flexibility is increased by Service-Oriented Architecture (SOA), which permits independent services like AuthService and ComparisonService to function independently. The MarketDataProvider uses the Adapter design to separate the system from external APIs. All things considered, these design decisions produce a modular, adaptable system that is simple to maintain and expand.

## Code Overview

### **Implementation**

The UserMgt Sys and application logic are implemented within the app package, using services such as ComparisonService, PriceManagementService, and SharePriceSystemService. These services handle core operations including comparison logic, price handling, and overall system coordination.

The PriceMgt Sys is implemented using the PriceRepository and the ExternalMarketAPIAdapter. The ExternalMarketAPIAdapter (located in the adapters.external package) connects to the external market API and converts the data into a format usable by the system. The PriceRepository acts as an intermediary, managing access to price data from both external and local sources.

The DataMgt Sys is implemented through data storage classes in the adapters.store package, including UserDataStore, PortfolioDataStore, and LocalPriceStore. These components handle persistent storage of user data, portfolio information, and locally cached share price data using JSON and CSV files.

The system also includes domain models such as ComparisonResult, which represent structured data used within the application. These models support clean data handling between components.

The Adapter pattern is clearly implemented through the ExternalMarketAPIAdapter, which isolates the system from the external API by adapting its interface to match the system's requirements.



| Component                | Code Class | Role                      |
|--------------------------|------------|---------------------------|
| ExternalMarketAPIAdapter | Adapter    | Connects to external API  |
| PriceRepository          | Repository | Manages price data        |
| LocalPriceStore          | Data store | Stores price data locally |
| UserDataStore            | Data store | Stores user data          |
| PortfolioDataStore       | Data store | Stores portfolio data     |
| ComparisonService        | Service    | Performs comparisons      |
| PriceManagementService   | Service    | Handles price logic       |

## Testing

| Test Case                            | Input                                 | Expected Result                              | Actual Result                 |
|--------------------------------------|---------------------------------------|----------------------------------------------|-------------------------------|
| Invalid login credentials            | Wrong username & password             | User cannot log in, error message shown      | Error message displayed       |
| Valid login credentials              | Correct username & password           | User successfully logs in                    | Login successful              |
| Compare output with valid date range | AAPL + MSFT, 2023-01-01 to 2023-06-01 | Graph shows both tickers over selected dates | Graph displayed correctly     |
| Invalid date selection               | End date before start date            | System rejects input and shows error         | Error message displayed       |
| Compare two companies                | AAPL + MSFT                           | Both companies displayed in graph            | Graph displayed correctly     |
| Add ticker to portfolio              | Logged in → add AAPL                  | Ticker is added to portfolio                 | Ticker added successfully     |
| View portfolio                       | Logged in → view portfolio            | Saved tickers are displayed                  | Portfolio displayed correctly |

**<https://github.com/HabsaSharif/Share-Price-Comparison.git>**

# Whole-System Architecture Principles in the Share Price App

For our project; objects model the business concepts, controllers handle user actions, services handle use cases, repositories coordinate data access, and adapters isolate the app from files and external APIs.

## Overall system shape

At a high level, the app follows this flow:

```
TerminalUI / Views
-> Controllers
-> Application Services
-> Repository / UserMgt services
-> Local stores or External API adapter
-> Domain objects passed back up for display
```

This matches the architecture diagram: UI compound component, ComparisonMgt, PriceMgt, UserMgt, DataMgt, plus an external SOA-style market API boundary.

## Quick code-to-architecture mapping

| Architecture part     | Code examples                                                                             | What it means                                                       |
|-----------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Domain layer          | Ticker, DateRange, PricePoint, PriceSeries, ComparisonResult, UserAccount, Portfolio      | Business concepts with minimal outside dependency.                  |
| Ports                 | IPriceMgt, IPriceRepository, IExternalAPI, IUserMgt, IPortfolioService                    | Interfaces that define what each component offers/requires.         |
| Application services  | ComparisonService, PriceManagementService, AuthService, PortfolioService                  | Use-case logic and coordination.                                    |
| Adapters / data layer | LocalPriceStore, PortfolioDataStore, UserDataStore, ExternalMarketAPIAdapter, ExternalAPI | File/API details kept away from the core use cases.                 |
| UI / MVC              | TerminalUI, LoginView, AuthController, SessionModel, ComparisonGraphView                  | User interaction, controller coordination, and view state updates.  |
| Bootstrap/wiring      | Application                                                                               | Builds the object graph and connects interfaces to implementations. |

## 1. Dependency Inversion Principle (DIP)

High-level logic usually depends on interfaces rather than concrete low-level classes.

```
private final IPriceMgt priceMgt;           // in ComparisonService
private final IPriceRepository repository; // in PriceManagementService
private final ILocalPriceStore localStore; // in PriceRepository
private final IExternalAPI externalApi;    // in PriceRepository
private final IUserMgt userMgt;           // in AuthController / PortfolioController
```

These important parts of the system are not hard-wired to a specific file store, API provider, or user-management implementation. They ask for an interface.

- ComparisonService does not know how prices are fetched. It only knows IPriceMgt.
- PriceManagementService does not know whether prices come from JSON, SQLite, or an API. It only knows IPriceRepository.
- PriceRepository does not know the concrete local store or concrete external API adapter. It only knows ILocalPriceStore and IExternalAPI.
- AuthController does not know the internals of AuthService, AccountService, or PortfolioService. It only knows IUserMgt.

This is useful because if we later replaced JSON storage with SQLite, or swapped the Stooq API for Alpha Vantage/Yahoo, the core flow doesn't need to be rewritten.

## 2. Single Responsibility Principle (SRP)

A lot of the classes have one clear job, which makes the code easier to explain and less chaotic.

```
Ticker           -> validates and represents a stock symbol
DateRange        -> validates and represents a time range
Portfolio        -> stores a user's unique saved tickers
ComparisonService -> builds a comparison result from one or two tickers
PriceRepository  -> decides local data first, external API second
ExternalAPI      -> performs the raw HTTP request
ExternalMarketAPIAdapter -> translates raw CSV into PriceSeries
TerminalUI       -> handles terminal menus and user input
```

This is good because the classes are not all doing everything. For example, ExternalAPI only knows HTTP and raw CSV. It does not know about PriceSeries. The adapter handles that translation.

## 3. Open/Closed Principle (OCP)

The app is reasonably open for extension without needing to modify existing classes. The interfaces make this possible.

```
class SQLitePriceStore implements ILocalPriceStore { ... }
class AlphaVantageAPIAdapter implements IExternalAPI { ... }
class SwingGraphView implements IViewPriceGraph { ... }
```

None of those would require changing the logic of ComparisonService or PriceManagementService.

- New data storage? Implement ILocalPriceStore.
- New market data provider? Implement IExternalAPI.
- New UI later? Keep the service interfaces and replace TerminalUI/views.

So basically the architecture is extendable.

## 4. Liskov Substitution Principle (LSP)

Any implementation of an interface is usable wherever the interface is expected, without the caller breaking.

Example: PriceRepository can use any IExternalAPI implementation.

```
PriceSeries fetched = externalApi.requestPrices(ticker, dateRange);
```

For LSP to work, every IExternalAPI implementation should return a valid PriceSeries for the requested ticker/date range, or fail with a pre-coded error. It shouldn't return random nulls, wrong tickers, or invalid price data.

The same applies to ILocalPriceStore. If LocalPriceStore is replaced with SQLitePriceStore, PriceRepository shouldn't care. It should still call read(...) and write(...) and get expected behaviour.

## 5. Interface Segregation Principle (ISP)

The project mostly uses small, focused interfaces rather than one giant interface that forces classes to implement things they do not need.

```
IAuthService      -> login/logout  
IAccountService   -> find account and check password  
IPortfolioService -> get portfolio and add ticker  
IExternalAPI      -> request prices  
ILocalPriceStore  -> read/write price series
```

That is better than having one huge IUserSystem interface with every account, portfolio, login, and price method crammed together.

## 6. Separation of Concerns

The project has a clear split between input/output, application logic, domain concepts, and external details.

```
UI / MVC  
    TerminalUI, LoginView, PortfolioView, ComparisonGraphView  
  
Application services  
    ComparisonService, PriceManagementService, AuthService, PortfolioService  
  
Domain  
    Ticker, DateRange, PriceSeries, Portfolio, UserAccount  
  
Adapters / external details  
    LocalPriceStore, UserDataStore, PortfolioDataStore, ExternalAPI, ExternalMarketAPIAdapter
```

This matters because changing how the app looks in the terminal should for example not force changes to how prices are fetched.

## 7. Ports and Adapters / Hexagonal Architecture

The code uses a Ports and Adapters style. The ports are the interfaces in the ports package. The adapters are the concrete classes that connect those ports to the outside world.

```
Ports:  
    IPriceRepository, IExternalAPI, ILocalPriceStore, IUserMgt, IPortfolioDataStore
```

Adapters:

```
PriceRepository, ExternalMarketAPIAdapter, ExternalAPI, LocalPriceStore, UserDataStore,
PortfolioDataStore
```

example is the market data path:

```
PriceRepository
-> IExternalAPI
-> ExternalMarketAPIAdapter
-> ExternalAPI
-> real web-service call
```

ExternalAPI deals with HTTP and the external service. ExternalMarketAPIAdapter converts raw CSV into PricePoint and PriceSeries. That means the rest of the system can stay in its own language: Ticker, DateRange, PriceSeries.

## 8. Layered / N-tier architecture

The code also has a layered shape, which matches the coursework diagrams.

Presentation tier:

```
TerminalUI and MVC views/controllers
```

Business/application tier:

```
ComparisonService, PriceManagementService, UserMgt services
```

Data and external tier:

```
LocalPriceStore, UserDataStore, PortfolioDataStore, ExternalAPI
```

example:

```
TerminalUI
-> ComparisonGraphView
-> ComparisonController
-> SharePriceSystemService / ComparisonService
-> PriceManagementService
-> PriceRepository
-> LocalPriceStore or ExternalMarketAPIAdapter
```

Each step moves from user interaction toward more specialised business/data logic.

## 9. MVC and publish/subscribe style in the UI

The UI has an MVC-style split. The View receives/display data, the Controller reacts to user actions, and the Model stores the current UI state.

```
LoginView -> AuthController -> SessionModel
PortfolioView -> PortfolioController -> SessionModel
ComparisonGraphView -> ComparisonController -> ComparisonModel
```

The publish/subscribe idea appears in the models. SessionModel stores observers and notifies them when login state changes:

```
private final List<SessionObserver> observers = new ArrayList<>();
```

```

public void subscribe(SessionObserver observer) {
    observers.add(observer);
}

private void notifyObservers() {
    for (SessionObserver observer : observers) {
        observer.onSessionChanged(this);
    }
}

```

When the session changes, the model tells interested views/controllers that the state changed. The model doesn't need to know exactly how each view will react.

## 11. Component-level principles and package cohesion

The packages mostly group classes that change for similar reasons:

```

domain -> business concepts
ports  -> interfaces / boundaries
app    -> application use cases
adapters -> files, repository, API integration
UI      -> terminal and MVC presentation
usermgmt -> account, login, portfolio services
bootstrap -> application wiring

```

This supports the Common Closure Principle: classes that are likely to change together are kept together. For example, changes to the external API mostly affect adapters/external. Changes to login/portfolio behaviour mostly affect usermgmt and UI/mvc.

It also supports the Common Reuse Principle reasonably well: if another part of the app uses domain objects, it is not forced to reuse UI classes or external API classes.



# Meeting Minutes

|                                                                           |                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                                      | 22/04/2026- 11:30am                                                                                                                                                                                                                                                              |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                                           |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Finalise system</li><li>• Complete report</li></ul>                                                                                                                                                                                      |
| <b>Facilitator</b>                                                        | Tobi                                                                                                                                                                                                                                                                             |
| <b>Note taker</b>                                                         | Habsa                                                                                                                                                                                                                                                                            |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                     |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa: Confirmed system is working</b></li><li>• <b>Tobi: Verified test cases and results</b></li><li>• <b>Sebastian: Reviewed final report formatting</b></li><li>• <b>Elijah: Checked report against marking criteria</b></li></ul> |
| <b>Discussion</b>                                                         | <ul style="list-style-type: none"><li>• Final review of system and report</li><li>• Ensured GitHub is updated</li></ul>                                                                                                                                                          |



|                                                       |                                                                                                               |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>points</b>                                         | <ul style="list-style-type: none"> <li>Confirmed readiness for submission</li> </ul>                          |
| <b>Actions (list tasks and assign a group member)</b> | <ul style="list-style-type: none"> <li>Upload final files – Team</li> <li>Submit coursework – Team</li> </ul> |



# Meeting Minutes

|                                                              |                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                         | 17/04/2026- 11:30am                                                                                                                                                                                                                              |
| <b>Project Name</b>                                          | Share-price-comparison                                                                                                                                                                                                                           |
| <b>Meeting Goal</b>                                          | <ul style="list-style-type: none"><li>• Finalise system</li><li>• Complete report</li></ul>                                                                                                                                                      |
| <b>Facilitator</b>                                           | Tobi                                                                                                                                                                                                                                             |
| <b>Note taker</b>                                            | sebastian                                                                                                                                                                                                                                        |
| <b>Attendees</b>                                             | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                     |
| <b>Roundtable Up-dates (each group member to contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa: Finalised system functionality</b></li><li>• <b>Tobi: Completed testing</b></li><li>• <b>Sebastian: Completed report structure</b></li><li>• <b>Elijah: Finalised report content</b></li></ul> |
| <b>Discussion</b>                                            | <ul style="list-style-type: none"><li>• Reviewed test results</li><li>• Ensured system meets all criteria</li></ul>                                                                                                                              |

|                                                       |                                                                                                                                                              |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>points</b>                                         | <ul style="list-style-type: none"> <li>• Checked report completeness</li> </ul>                                                                              |
| <b>Actions (list tasks and assign a group member)</b> | <ul style="list-style-type: none"> <li>• Final system review – Team</li> <li>• Proofread report – Team</li> <li>• Prepare submission files – Team</li> </ul> |



# Meeting Minutes

|                                                                           |                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                                      | 10/04/2026- 11:30am                                                                                                                                                                                                                                          |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                       |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Begin testing phase</li><li>• Review completed features</li></ul>                                                                                                                                                    |
| <b>Facilitator</b>                                                        | Habsa                                                                                                                                                                                                                                                        |
| <b>Note taker</b>                                                         | elijah                                                                                                                                                                                                                                                       |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                 |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa: Completed most coding tasks</b></li><li>• <b>Tobi: Started writing test cases</b></li><li>• <b>Sebastian: Continued report writing</b></li><li>• <b>Elijah: Worked on implementation section</b></li></ul> |
| <b>Discussion<br/>points</b>                                              | <ul style="list-style-type: none"><li>• Identified test scenarios (valid/invalid inputs)</li><li>• Reviewed system functionality</li><li>• Checked alignment with requirements</li></ul>                                                                     |

**Actions (list  
tasks and assign  
a group member)**

- Complete test cases – Tobi
- Fix bugs from testing – Team
- Continue report writing – Elijah & Sebastian



# Meeting Minutes

|                                                                           |                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                                      | 03/04/2026- 11:30am                                                                                                                                                                                                                                                    |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                                 |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Progress implementation</li><li>• Introduce SOA concepts</li></ul>                                                                                                                                                             |
| <b>Facilitator</b>                                                        | Habsa                                                                                                                                                                                                                                                                  |
| <b>Note taker</b>                                                         | elijah                                                                                                                                                                                                                                                                 |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                           |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa: Continued coding system logic</b></li><li>• <b>Tobi: Assisted with component connections</b></li><li>• <b>Sebastian: Documented architecture styles</b></li><li>• <b>Elijah: Worked on SOA explanation</b></li></ul> |
| <b>Discussion<br/>points</b>                                              | <ul style="list-style-type: none"><li>• Introduced SOA principles (modularity, interoperability)</li><li>• Ensured components follow design patterns</li><li>• Identified missing features</li></ul>                                                                   |

**Actions (list  
tasks and assign  
a group member)**

- Implement remaining components – Team
- Add SOA section to report – Elijah
- Review architecture consistency – Sebastian



# Meeting Minutes

|                                                                           |                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                                      | 27/03/2026- 11:30am                                                                                                                                                                                                                                                    |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                                 |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Review coding progress</li><li>• Continue implementation</li></ul>                                                                                                                                                             |
| <b>Facilitator</b>                                                        | Habsa                                                                                                                                                                                                                                                                  |
| <b>Note taker</b>                                                         | elijah                                                                                                                                                                                                                                                                 |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                           |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa: Implemented core system structure</b></li><li>• <b>Tobi: Worked on integrating components</b></li><li>• <b>Sebastian: Began writing architecture section</b></li><li>• <b>Elijah: Started report draft</b></li></ul> |
| <b>Discussion<br/>points</b>                                              | <ul style="list-style-type: none"><li>• Reviewed component structure</li><li>• Ensured alignment with architecture styles</li><li>• Discussed reusable component design</li></ul>                                                                                      |



**Actions (list  
tasks and assign  
a group member)**

- Continue component implementation – Team
- Expand report (architecture explanation) – Elijah & Sebastian
- Push updates to GitHub – Team



# Meeting Minutes

|                                                                           |                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                                      | 20/03/2026- 11:30am                                                                                                                                                                                                                                                               |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                                            |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Kick off Sprint 3</li><li>• Review brief and assign tasks</li></ul>                                                                                                                                                                       |
| <b>Facilitator</b>                                                        | Tobi                                                                                                                                                                                                                                                                              |
| <b>Note taker</b>                                                         | Habsa                                                                                                                                                                                                                                                                             |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                      |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa:</b> Discussed coding approach and GitHub setup</li><li>• <b>Tobi:</b> Reviewed Sprint 2 outputs for reuse</li><li>• <b>Sebastian:</b> Reviewed report expectations</li><li>• <b>Elijah:</b> Analysed marking criteria</li></ul> |
| <b>Discussion<br/>points</b>                                              | <ul style="list-style-type: none"><li>• Reviewed Sprint 3 brief (compound components + architecture styles)</li><li>• Agreed on layered/MVC approach</li></ul>                                                                                                                    |

|                                                       |                                                                                                                                                                                                             |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | <ul style="list-style-type: none"> <li>• Identified key components to implement</li> <li>• Confirmed team roles</li> </ul>                                                                                  |
| <b>Actions (list tasks and assign a group member)</b> | <ul style="list-style-type: none"> <li>• Set up project structure – Habsa</li> <li>• Begin Price Management component – Team</li> <li>• Start report outline – Elijah &amp; Sebastian &amp; Tobi</li> </ul> |



# Meeting Minutes

|                                                             |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                        | 13/03/2026- 11:30am                                                                                                                                                                                                                                                                                                            |
| <b>Project Name</b>                                         | Share-price-comparison                                                                                                                                                                                                                                                                                                         |
| <b>Meeting Goal</b>                                         | <ul style="list-style-type: none"><li>• Final review of Sprint 2 work</li><li>• Prepare for submission</li></ul>                                                                                                                                                                                                               |
| <b>Facilitator</b>                                          | Tobi                                                                                                                                                                                                                                                                                                                           |
| <b>Note taker</b>                                           | Habsa                                                                                                                                                                                                                                                                                                                          |
| <b>Attendees</b>                                            | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                                                                   |
| <b>Roundtable Updates (each group member to contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa:</b> Ensured all files uploaded and GitHub is updated</li><li>• <b>Elijah:</b> Completed Business Concept Model and Use Cases</li><li>• <b>Tobi:</b> Finalised System Interfaces</li><li>• <b>Sebastian:</b> Completed Initial Architecture and Business Interfaces</li></ul> |
| <b>Discussion points</b>                                    | <ul style="list-style-type: none"><li>• Conducted full review of all Sprint 2 deliverables</li><li>• Checked consistency across diagrams and explanations</li><li>• Ensured coursework requirements are met</li><li>• Confirmed readiness for submission</li></ul>                                                             |

**Actions (list  
tasks and assign  
a group member)**

- Submit Sprint 2 work – Team
- Final GitHub check – Habsa
- Proofread and formatting check – All members



# Meeting Minutes

|                                                                           |                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                                      | 06/03/2026- 11:30am                                                                                                                                                                                                                                                                                                       |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                                                                                    |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Continue development</li><li>• Align all sections</li></ul>                                                                                                                                                                                                                       |
| <b>Facilitator</b>                                                        | Tobi                                                                                                                                                                                                                                                                                                                      |
| <b>Note taker</b>                                                         | Habsa                                                                                                                                                                                                                                                                                                                     |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                                                              |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa:</b> Maintaining GitHub and tracking progress</li><li>• <b>Elijah:</b> Refining Business Concept Model and Use Cases</li><li>• <b>Tobi:</b> Improving System Interfaces diagrams</li><li>• <b>Sebastian:</b> Updating architecture explanations and interfaces</li></ul> |
| <b>Discussion<br/>points</b>                                              | <ul style="list-style-type: none"><li>• Checked alignment between models and architecture</li><li>• Reviewed feedback within the team</li><li>• Discussed improving explanations in documentation</li><li>• Confirmed progress is on track</li></ul>                                                                      |

**Actions (list  
tasks and assign  
a group member)**

- Finalise individual sections – All members
- Review and improve documentation quality – Team
- Prepare for final review meeting – All members



# Meeting Minutes

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                        | 27/02/2026- 4:00pm                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Project Name</b>                                         | Share-price-comparison                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Meeting Goal</b>                                         | <ul style="list-style-type: none"><li>• Review progress of Sprint 2 tasks</li><li>• Identify any issues or blockers</li><li>• Ensure all components are developing correctly</li></ul>                                                                                                                                                                                                                          |
| <b>Facilitator</b>                                          | Elijah                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Note taker</b>                                           | Sebastian                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Attendees</b>                                            | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                                                                                                                                                    |
| <b>Roundtable Updates (each group member to contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa: Progressing with GitHub and repository organisation</b></li><li>• <b>Elijah: Drafted Business Concept Model and Use Cases</b></li><li>• <b>Tobi: Working on System Interfaces diagrams</b></li><li>• <b>Sebastian: Drafted Initial Architecture and Business Interfaces</b></li><li>• <b>Team: Reviewed each other's work and provided feedback</b></li></ul> |
| <b>Discussion</b>                                           | <ul style="list-style-type: none"><li>• Reviewed initial drafts</li><li>• Identified inconsistencies between diagrams</li></ul>                                                                                                                                                                                                                                                                                 |



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| <b>points</b>                                         | <ul style="list-style-type: none"> <li>• Discussed improving clarity and structure</li> <li>• Agreed to refine work before next meeting</li> </ul>                                                                 |
| <b>Actions (list tasks and assign a group member)</b> | <ul style="list-style-type: none"> <li>• Refine diagrams and content – <b>All members</b></li> <li>• Ensure consistency across sections – <b>Team</b></li> <li>• Continue GitHub updates – <b>Habsa</b></li> </ul> |



# Meeting Minutes

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                        | 19/02/2026- 4:00pm                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Project Name</b>                                         | Share-price-comparison                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Meeting Goal</b>                                         | <ul style="list-style-type: none"><li>• Review progress of Sprint 2 tasks</li><li>• Identify any issues or blockers</li><li>• Ensure all components are developing correctly</li></ul>                                                                                                                                                                                                                                                                    |
| <b>Facilitator</b>                                          | Elijah                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Note taker</b>                                           | Sebastian                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Attendees</b>                                            | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                                                                                                                                                                                              |
| <b>Roundtable Updates (each group member to contribute)</b> | <ul style="list-style-type: none"><li>• <b>Habsa:</b> Progressing with GitHub updates and repository management</li><li>• <b>Elijah:</b> Completed draft of Business Concept Model and Use Case Model</li><li>• <b>Tobi:</b> Working on System Interfaces and refining diagrams</li><li>• <b>Sebastian:</b> Drafted Initial System Architecture and Business Interfaces</li><li>• <b>Team:</b> Reviewed each other's work and provided feedback</li></ul> |

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| <b>Discussion points</b>                              | <ul style="list-style-type: none"> <li>• Reviewed current progress of all Sprint 2 components</li> <li>• Identified minor inconsistencies between diagrams and descriptions</li> <li>• Discussed improvements to clarity and structure of architecture diagrams</li> <li>• Agreed to refine and align all sections for consistency</li> <li>• Emphasised need to complete all sections before final review</li> </ul> |
| <b>Actions (list tasks and assign a group member)</b> | <ul style="list-style-type: none"> <li>• Refine diagrams and ensure consistency – Tobi &amp; Sebastian</li> <li>• Finalise Business Concept Model and Use Cases – Elijah</li> <li>• Continue GitHub updates and organise files – Habsa</li> <li>• Review all sections for clarity and accuracy – All members</li> <li>• Prepare for final Sprint 2 review – Team</li> </ul>                                           |



# Meeting Minutes

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| <b>Date and Time</b>                                                      | 17/02/2026- 11:30                                                                                                                                                                                                                                                                                                            |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                                                                                       |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Finalise Sprint 1 Work.</li><li>• Prepare for submission.</li></ul>                                                                                                                                                                                                                  |
| <b>Facilitator</b>                                                        | Elijah                                                                                                                                                                                                                                                                                                                       |
| <b>Note taker</b>                                                         | Sebstatian                                                                                                                                                                                                                                                                                                                   |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                                                                 |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Sebastian:</b> Completed component diagram</li><li>• <b>Elijah/Tobi:</b> Completed code of conduct (including penalties)/completed documentation</li><li>• <b>Habsa:</b> Finalised repository setup</li><li>• <b>Sebstatian/Elijah:</b> Reviewed architecture decisions</li></ul> |

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| <b>Discussion points</b>                              | <ul style="list-style-type: none"> <li>• Reviewed architectural design document</li> <li>• Ensured all Sprint 1 deliverables completed</li> <li>• Prepared for submission deadline (19/02/2026)</li> </ul> |
| <b>Actions (list tasks and assign a group member)</b> | <ul style="list-style-type: none"> <li>• Submit architecture document – <i>Team</i></li> <li>• Final review of GitHub – <i>Habsa</i></li> <li>• Ensure all files uploaded – <i>Team</i></li> </ul>         |



# Meeting Minutes

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|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date and Time</b>                                                      | 12/02/2026- 11:30                                                                                                                                                                                                                                                                                          |
| <b>Project Name</b>                                                       | Share-price-comparison                                                                                                                                                                                                                                                                                     |
| <b>Meeting Goal</b>                                                       | <ul style="list-style-type: none"><li>• Develop System design</li><li>• Define architecture componenets</li></ul>                                                                                                                                                                                          |
| <b>Facilitator</b>                                                        | Sebastian                                                                                                                                                                                                                                                                                                  |
| <b>Note taker</b>                                                         | Tobi                                                                                                                                                                                                                                                                                                       |
| <b>Attendees</b>                                                          | <ul style="list-style-type: none"><li>• <b>Habsa Sharif</b></li><li>• <b>Tobi Obafemi</b></li><li>• <b>Seb Moreno</b></li><li>• <b>Elijah Kuku</b></li></ul>                                                                                                                                               |
| <b>Roundtable Up-<br/>dates (each<br/>group member to<br/>contribute)</b> | <ul style="list-style-type: none"><li>• <b>Sebastian:</b> Worked on component diagram</li><li>• <b>Elijah/Tobi:</b> Continued code of conduct/ Updated meeting documentation</li><li>• <b>Habsa:</b> Organised repository structure</li><li>• <b>Sebastian/Habsa:</b> Explored API/data handling</li></ul> |
| <b>Discussion<br/>points</b>                                              | <ul style="list-style-type: none"><li>• Defined system components:</li></ul>                                                                                                                                                                                                                               |

|                                                       |                                                                                                                                                                                                                                                            |
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|                                                       | <ul style="list-style-type: none"> <li>○ User Interface (UI)</li> <li>○ Comparison Service</li> <li>○ Price Repository</li> <li>○ External API</li> <li>• Discussed separation of layers</li> </ul>                                                        |
| <b>Actions (list tasks and assign a group member)</b> | <ul style="list-style-type: none"> <li>• Complete component diagram – <i>Sebastian</i></li> <li>• Finalise code of conduct – <i>Elijah</i></li> <li>• Setup Java project structure – <i>Habsa</i></li> <li>• Plan API interaction – <i>Tobi</i></li> </ul> |